^{17}O NMR study of the inhomogeneous electronic state in $La_{2-x}Sr_xCuO_4$ crystals

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We report measurements on the inhomogeneous line broadening of the ^{17}O NMR spectrum in the high T_c superconductor $La_{2-x}Sr_xCuO_4$ for single crystals with nominal hole concentrations $x=0.035$, 0.05, 0.07, 0.115, and 0.15. A substantial overlap between the ^{17}O NMR spectrum of different crystals is reported, and the extent of the overlap is shown to increase with decreasing temperature. The spin-lattice relaxation rate divided by temperature, $1/^{17}T_1T$, is found to vary linearly with frequency (or Knight shift $^{17}K_{spin}$, equivalently) across the ^{17}O NMR spectrum of *each* crystal, with the *same* scaling form given by $1/^{17}T_1T=\beta^{17}K_{spin}$, where $\beta=2.4(2)$ $(\mathrm{sK}\%)^{-1}$. We therefore argue that the overlap between the ^{17}O NMR spectrum of different crystals x is a result of an inherent spatial variation in hole concentration within each crystal, and that the extent of the line broadening, $^{17}\Delta K_{spin}$, is a measure of the extent $^{17}\Delta x_{local}$ (or amplitude) of the spatial variation in *local* hole concentration, $^{17}x_{local}=x\pm{}^{17}\Delta x_{local}$. We successfully fit the ^{17}O NMR line shapes with a simple numerical model that determines the random distribution in local hole concentration $^{17}x_{local}$ as a result of the random spatial distribution in Sr^{2+} hole donors within a patch radius $^{17}R_{patch}$. We find that $^{17}R_{patch}=2-5$ nm best fits the data, implying a short length scale for the inhomogeneous electronic state. In this report we present the first systematic study of the spatial distribution in the local static spin susceptibility in the high T_c cuprates.

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I. INTRODUCTION

There is increasing evidence from NMR (nuclear magnetic resonance) and NQR (nuclear quadrupole resonance) measurements[1–13] that there is some sort of inhomogeneity in the CuO_2 planes of $La_{2-x}[Sr,Ba]_xCuO_4$, its Nd or Eu codoped compound $[La,Nd,Eu]_{2-x}[Sr,Ba]_xCuO_4$ and also superoxygenated $La_2CuO_{4+\delta}$. Over the past several years, studies using ^{63}Cu NQR and NMR wipeout[6–8,10,11] have characterized the glassy nature of the slowing down of the stripe[14] inhomogeneity in $[La,Nd,Eu]_{2-x}[Sr,Ba]_xCuO_4$, where the Coulomb potential from the distorted lattice slows down spin and charge density waves.[14] It has also been shown that the high mobility of the oxygen atoms in $La_2CuO_{4+\delta}$ results in electronic phase separation[15–17] between the superconducting and antiferromagnetic phase, as evidenced by ^{139}La NMR.[3,4] More recently, ^{63}Cu NQR wipeout[18] and ^{89}Y NMR[18,19] in superconducting $Y_{1-z}Ca_zBa_2Cu_3O_6$ have revealed the presence of slowing inhomogeneities.

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No clear picture has emerged that discerns and relates the effects of genuine electronic phase separation,[15–17] stripe modulations,[14] chemical inhomogeneities due to ionic radius mismatches,[20] and, electronic inhomogeneities due to Coulomb potentials from the donor ions. STM (scanning tunnelling microscopy) studies on the surface state of $Bi_2Sr_2CaCu_2O_{8+\delta}$ cleaved at low temperature[21] have revealed spatial variations of the electronic state on a ~nm scale in the surface plane. More recently, STM studies on lightly doped $Ca_{2-x}Na_xCuO_2Cl_2$ (Ref. 22) have revealed similar ~nm inhomogeneities of the electronic state. Whether such ~nm modulations are universally observable in the bulk and other high T_c materials remains to be seen. One counterexample is superconducting $YBa_2Cu_3O_{6+\delta}$, where the inhomogeneous line broadening of the ^{89}Y NMR spectra[23] indicates that hole doping in $YBa_2Cu_3O_{6+\delta}$ is homogeneous in comparison. It has been commented, however, that a large but commensurate hole density variation cannot be excluded from the ^{89}Y NMR data alone.[24]

Using ^{63}Cu NQR measurements[12,25] on ^{63}Cu isotope-enriched $La_{2-x}Sr_xCuO_4$ polycrystalline powders, we previously reported evidence for a substantial *local* variation in hole concentration $^{63}x_{local}$ away from its nominal value x by measuring the frequency dependence of the spin-lattice relaxation rate $1/^{63}T_1$[5,12] across the inhomogeneously broadened ^{63}Cu NQR spectrum. The ^{63}Cu NQR spectrum is inhomogeneously broadened by spatial variations in the electric field gradient at the Cu nuclear site, and therefore the extent of the intrinsic frequency dependence of $1/^{63}T_1$ across the NQR spectrum is determined through the *charge* channel. In this paper we report phenomena through the *spin* channel, which we can measure using the inhomogeneously broadened planar ^{17}O NMR spectrum in high-quality $La_{2-x}Sr_xCuO_4$ single crystals.[26]

The key new ^{17}O NMR results are presented in Fig. 1, where at 295 K we present in (a) the frequency dependence, or equivalently the apparent Knight shift ($^{17}K^c_{app}$) dependence, of the spin-lattice relaxation rate divided by temperature T, $1/^{17}T_1T^c$, of the planar ^{17}O central transition spectrum shown in part (b) (where the superscript c denotes that the external magnetic field is aligned along the crystal c axis). Figure 1(a) clearly shows that the empirical scaling relation,

$$\frac{1}{^{17}T_1T^c} = \beta\,^{17}K^c_{spin}, \tag{1}$$

holds *quantitatively* across *each* spectrum with slope $\beta = 2.4(2)$ $(\mathrm{sK\%})^{-1}$. We recall that in $YBa_2Cu_3O_{6.6}$, where the ^{17}O NMR linewidth is a factor ~2 smaller than our $x = 0.15$ crystal at 100 K, Takigawa *et al.*[27] found the *same* universal scaling phenomenology.[28,29] In Ref. 27 however, β^{CG} was obtained by plotting $\langle 1/^{17}T_1T^c\rangle$ against $\langle ^{17}K^c_{spin}\rangle$ *as a function of temperature* and at the CG (center of gravity), i.e., the mean position, of the ^{17}O NMR spectrum, and a consistent value $\beta^{CG} \sim 2.0(2)$ $(\mathrm{sK\%})^{-1}$ was reported.

The universal scaling relation in Eq. (1) across each spectrum of $La_{2-x}Sr_xCuO_4$ implies that each Knight shift value corresponds to a unique local electronic state with its characteristic value of $1/^{17}T_1T^c$. If, instead, the ^{17}O NMR line broadening were dominated by some kind of "dirt" effect (similar to the effects of localized spins induced by Zn^{2+} impurities), one would not observe the correlation between the Knight shift and $1/^{17}T_1T^c$ given in Eq. (1), with the *same* value of β found across all the $La_{2-x}Sr_xCuO_4$ spectra over a wide temperature range (see later), and, furthermore, with the same β as found in $YBa_2Cu_3O_{6+\delta}$. Given these findings and observations, we can then use the line broadening and overlap of the ^{17}O NMR spectrum for different nominal x [Fig. 1(b)] to deduce the local electronic state in each crystal. For example, the upper (lower) half intensity point of the $x = 0.05$ crystal approximately coincides with the CG (center of gravity) of the $x = 0.07$ (0.035) crystal, respectively, which indicates that at 295 K the more metallic regions of the $x = 0.05$ crystal have local hole concentrations $^{17}x_{local} = 0.07$, while the more insulating regions have $^{17}x_{local} = 0.035$. This implies that for $x = 0.05$ at 295 K the characteristic extent, or

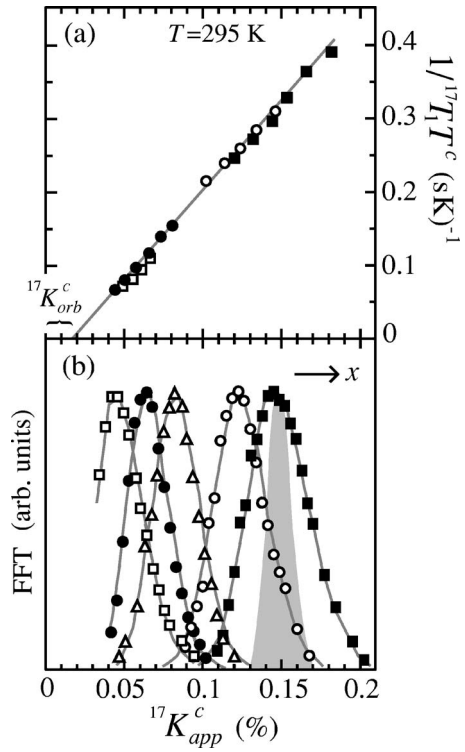


FIG. 1. (a) Frequency dependence of the planar ^{17}O NMR spin-lattice relaxation rate divided by temperature T, $1/^{17}T_1T^c$ $(\mathrm{sK})^{-1}$, across the NMR spectrum presented in (b). The solid line in (a) is best fit to Eq. (1) with slope $\beta = 2.4$ $(\mathrm{sK\%})^{-1}$, and the K-axis offset defines constant orbital shift $^{17}K^c_{orb} = +0.018\%$, where $^{17}K^c_{app} = {}^{17}K^c_{orb} + {}^{17}K^c_{spin}$. (b) The planar ^{17}O FFT (fast Fourier transform) spectrum (arbitrary units) at the central transition as a function of $^{17}K^c_{app}$ for the c-axis aligned $La_{2-x}Sr_xCuO_4$ single crystals with $x = 0.15$ (■), $x = 0.115$ (○), $x = 0.07$ (△), $x = 0.05$ (●), and $x = 0.035$ (□). All data are taken at a fixed temperature of 295 K. Missing data points for $x = 0.035$ and 0.05 in (b) indicate a region where the apical oxygen signal coincides. Solid lines in (b) are best fits using a patch model for the spatial variation $^{17}x_{local}$, with a patch radius of $^{17}R_{patch} = 3.0-3.5$ nm in order of increasing x, respectively. The shaded line shape shows line broadening according to the EPMA for $x = 0.15$ (see below).

amplitude, of the spatial inhomogeneity is $^{17}\Delta x_{local}=0.018(2)$, where we define $^{17}\Delta x_{local}$ through the following relation:

$$^{17}x_{local} = x \pm {}^{17}\Delta x_{local}. \quad (2)$$

Furthermore, we find that the scaling relation shown in Fig. 1(a) at 295 K holds quantitatively at different temperatures within the temperature range $T_{scale} \leqslant T \leqslant 475$ K, where $T_{scale} \geqslant 100$ K. Below T_{scale}, the scaling breaks down, implying that Eq. (2) is no longer valid, and therefore we limit our determination of $^{17}x_{local}$ above T_{scale}.

The rest of this paper is sectioned as follows: in Sec. II we discuss the sample characteristics and the NMR background. In Sec. III we present the experimental results of the Knight shift and spin-lattice relaxation rate, and we determine the temperature dependence of the amplitude of the spatial variation in electronic states. In Sec. IV we present the length scale analysis to fit the data. We present our conclusions in Sec. V.

II. EXPERIMENTAL DETAILS

A. Samples

We used high-quality single crystals of $La_{2-x}Sr_xCuO_4$ with nominal hole concentrations of $x=0.025$, 0.035, 0.05, 0.07, 0.115, and 0.15 with masses ranging from 100–500 mg for our ^{17}O NMR measurements. The crystals were grown at MIT ($x=0.025, 0.035, 0.07$), at Tohoku ($x=0.115$) and Tokyo ($x=0.05$ and 0.15) using an optical image furnace. ^{17}O isotope enrichment was carried out by annealing the crystals in $^{17}O_2$ enriched gas for 100 h at 900 C, and were aligned along the crystal c axis and $a(b)$ axis. The single crystals were not taken over 475 K in order to ensure that the oxygen content did not alter from one experimental run to another.

EPMA (electron probe microanalysis) on the $x=0.15$ crystal showed that the spatial variation of the Sr^{2+} content averaged over the focus area of the electron beam of ~ 1 μm is $\Delta x_{epma} \simeq 0.012$. This is consistent with earlier works in $La_{2-x}Sr_xCuO_4$ powders, where the width of the high resolution x-ray diffraction lines[30,31] was used to determine the variations in Sr^{2+} content over length scales $\geqslant 10$ nm. They found a consistent value $\Delta x \simeq 0.01$ that they attribute to imperfect mixing in the reaction process.[30] The x-ray findings, together with our EPMA measurement suggest that the mean Sr^{2+} content varies over macroscopic ($>\mu$m) length scales in the crystal as a result of imperfect mixing, which is independent of temperature. The EPMA value of $\Delta x_{epma} \simeq 0.012$ for $x=0.15$ therefore results in a temperature-independent linewidth (see Sec. III) shown by the shaded line shape in Fig. 1(b), which is small compared to the line broadening from the local spatial variation in hole concentration $^{17}\Delta x_{local}$. In other words, we can immediately distinguish between the temperature-independent variation Δx_{epma}, which modulates over macroscopic ($>\mu$m) length scales, to the temperature-dependent spatial variation in the local hole concentration $^{17}\Delta x_{local} (\gg \Delta x_{epma})$, which modulates over microscopic ($\sim$nm) length scales (Sec. IV).

B. NMR background

^{17}O is a spin $I=5/2$ nucleus, which, in the presence of high applied magnetic fields, splits into a total of $2I=5$ lines through the nuclear quadrupole interaction. The optimum configuration for the purposes of this report was to use the central transition with the applied magnetic field aligned along the crystal c axis. The reason for this is that the central transition, which corresponds to the transitions $(-1/2 \leftrightarrow +1/2)$ between the magnetic eigenstates $(m_I \leftrightarrow m_I')$, is broadened by local distributions in the uniform static spin susceptibility, and has no appreciable broadening due to charge effects (i.e. no quadrupole broadening).[32,33] We then choose to orient the magnetic field along the crystal c axis so as to avoid a coincidence with the much narrower apical ^{17}O NMR spectrum.

The planar ^{17}O NMR spectrum was obtained by taking the fast Fourier transform (FFT) of the spin echo. The $\pi/2$ pulse length was chosen so as to fully excite the central transition while not disturbing the quadrupole split satellites $\sim \pm 200$ kHz away. The NMR frequency of the central transition along the c axis is given by[32,33]

$$^{17}\nu(x,T)^c = (1 + {}^{17}K^c_{orb} + {}^{17}K^c_{spin}(x,T))\,\nu_0 + {}^{17}\nu^c_{Q(2)}(x), \quad (3)$$

where $\nu_0 = {}^{17}\gamma H_0/2\pi = 51.948$ MHz is the bare resonance frequency in the applied magnetic field of $H_0=9$ T. The field was calibrated using the D resonance in D_2O with an observed spectral width of ~ 5 ppm, which corresponds to the delta function resolution in Fig. 1. $^{17}\nu^c_{Q(2)}$ in Eq. (3) is the second-order quadrupole shift of the central transition. By measuring the mean values of the principle components of the quadrupole tensor (the values of which were consistent with previous works[34]), we deduce the small second-order shifts $^{17}\nu^c_{Q(2)} \leqslant 5$ kHz to the central transitions.[33] All NMR spectra presented here have the appropriate shifts $^{17}\nu^c_{Q(2)}/\nu_0$ ($\sim +0.01\%$ equivalently) already subtracted. $^{17}K^c_{org}$ in Eq. (3) is the single valued, x- and T-independent orbital (Van-Vleck) shift.[35] We deduce that $^{17}K^c_{org}$ by the zero intercept of the scaling relation, as illustrated in Fig. 1, which yields $^{17}K^c_{org} = +0.018(5)\%$, in agreement with previous estimates.[35]

The remaining shift, $^{17}K^c_{spin}$ in Eq. (3), is the distributed spin shift in which we are interested. The standard expression for $^{17}K^c_{spin}$ is given by

$$^{17}K^c_{spin} = \frac{2C^c}{N_A\mu_B}\chi'^c_{spin}(\mathbf{q}=\mathbf{0}), \quad (4)$$

where χ'^c_{spin} is the static uniform $(\mathbf{q}=\mathbf{0})$ molar spin susceptibility per formula unit along the c axis, and C^c is the c-axis hyperfine coupling constant from each of the two Cu neighbors. Defining $\parallel$ and $\perp$ as the two in-plane crystal axes parallel and perpendicular to the Cu—O—Cu bond, respectively, we find the anisotropic hyperfine coupling constants to be $[2C^{\parallel}, 2C^{\perp}, 2C^c] = [143(10), 90(8), 88(8)]$ (kOe/μ_B) according to $K-\chi$ analysis, the results of which are consistent with previous estimates.[36–41]

The spin-lattice relaxation rate, $1/{}^{17}T^c_1$, was measured by applying an additional π inversion pulse at a delay-time t prior to the spin-echo sequence. The value of $1/{}^{17}T^c_1$ was

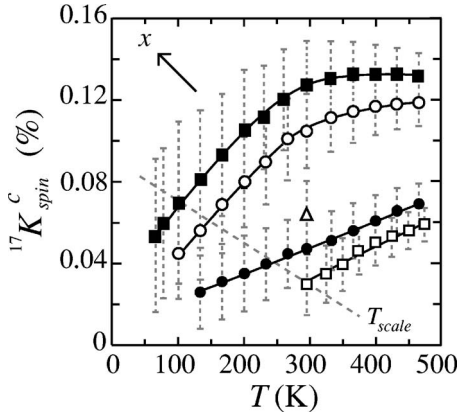


FIG. 2. Temperature dependence of the spin shift $^{17}K^{c}_{spin}$ (%) for $x=0.15$ (■), $x=0.115$ (○), $x=0.07$ (△), $x=0.05$ (●) and $x=0.035$ (□). Data points indicate the CG (center of gravity) of the spectrum, $\langle {}^{17}K^{c}_{spin}\rangle$, while error bars indicate ± HWHM (half-width at half-maximum), $\pm {}^{17}\Delta K^{c}_{spin}$. Black curves are guides for the eye, and a dashed grey line shows T_{scale}.

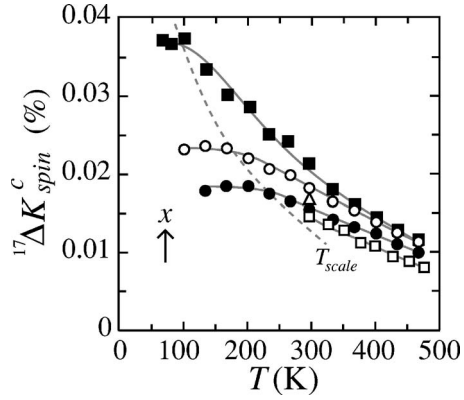


FIG. 3. Temperature dependence of the HWHM, $^{17}\Delta K^{c}_{spin}$ (%), for $x=0.15$ (■), $x=0.115$ (○), $x=0.07$ (△), $x=0.05$ (●), and $x=0.035$ (□). Solid lines are a guide for the eye and a dashed curve shows T_{scale}.

deduced by fitting the signal intensity of the FFT spectrum, for various inversion times t, to the standard expression for the recovery of the central transition $R(t)$.[43] In the case where $1/{}^{17}T_1$ is independent of frequency, the integral of the entire FFT spectrum is taken, and a fit to $R(t)$ results in the single value $1/{}^{17}T_1$. However, in this paper, $1/{}^{17}T_1$ is frequency (or shift) dependent and therefore a fit to $R(t)$ must be taken at each desired shift along the FFT spectrum, resulting in a different value of $1/{}^{17}T_1$ at each shift. $R(t)$ at *each* shift fit well to the standard recovery $R(t)$,[43] implying a single value of $^{17}1/T_1$ at each shift. In the case of the $x=0.035$ crystal, and to a lesser extent $x=0.05$, interference with the apical oxygen site at lower shifts resulted in an additional $\gtrsim 10$ times longer $^{17}T_1$ component, thereby disabling our procedure. The missing planar oxygen spectra and $1/{}^{17}T_1$ data for $x=0.035$ and 0.05 in Fig. 1 indicate where coincidence with the apical oxygen occurs.

In the present case, the spin-lattice relaxation rate divided by temperature T, $1/{}^{17}T_1T^c$, is dominated by local magnetic fluctuations, and is given by the standard expression[44]

$$\frac{1}{{}^{17}T_1T^c} = \frac{{}^{17}\gamma^2 k_B}{g^2\mu_B^2}\sum_{\mathbf{q},\alpha} |{}^{17}A(\mathbf{q})^{\alpha}|^2 \frac{\chi''^{\alpha}_{spin}(\mathbf{q},\omega_0)}{\omega_0},$$

$$ {}^{17}A(\mathbf{q})^{\alpha} = 2C^{\alpha}\cos\left(\frac{q_{\alpha}a}{2}\right), \tag{5}$$

for c-axis orientation, where $\alpha=(\parallel,\perp)$. $\chi''^{\alpha}_{spin}(\mathbf{q},\omega_0)$ is the imaginary part of the dynamic spin susceptibility, $^{17}A(\mathbf{q})^{\alpha}$ is the $\mathbf{q}$-dependent anisotropic hyperfine coupling, and $a \simeq 0.38$ nm is the in-plane lattice constant in the tetragonal phase.

III. EXPERIMENTAL RESULTS

A. Experimental data

In Fig. 2 we present the temperature dependence of the spin Knight shift $^{17}K^{c}_{spin}$ for the series of samples. The data points in Fig. 2 correspond to the CG (center of gravity or mean position), $\langle {}^{17}K^{c}_{spin}\rangle$, while the error bars are the HWHM (half-width at half-maximum), $^{17}\Delta K^{c}_{spin}=\sqrt{2Ln2}\sigma_K$, where σ_K is the square root of the calculated second moment.[42] We choose to plot the data in this fashion in order to illustrate the increasing spectral overlap (and therefore of $^{17}\Delta x_{local}$) with decreasing temperature. In Fig. 2 we also show the temperature scale T_{scale} (dashed line) below which the scaling relation in Eq. (1) no longer holds.

The temperature dependence of $^{17}\Delta K^{c}_{spin}$ is shown in Fig. 3. We find a monotonic increase of the HWHM $^{17}\Delta K^{c}_{spin}$ from 475 K down to $\sim T_{scale}$, below which $^{17}\Delta K^{c}_{spin}$ *saturates*. We note that the fact that $^{17}\Delta K^{c}_{spin}$ saturates at low temperatures is further evidence that the line broadening is not caused by a simple Curie-type mechanism, as previously speculated. We also note that even though our single crystal $x=0.15$ linewidth at 295 K is ~2 narrower than previous

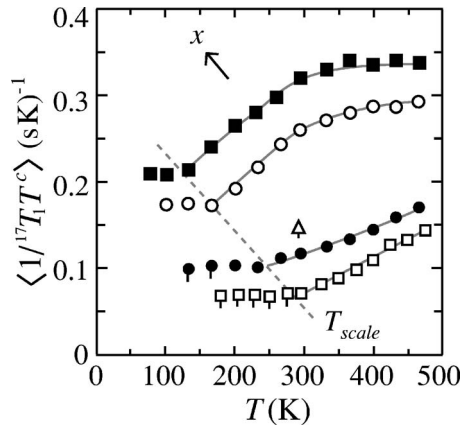


FIG. 4. Temperature dependence of $\langle 1/{}^{17}T_1T^c\rangle$ (sK)$^{-1}$ at the CG of the central transition for $x=0.15$ (■), $x=0.115$ (○), $x=0.07$ (△), $x=0.05$ (●), and $x=0.035$ (□). Symbols with lines underneath indicate $\langle 1/{}^{17}T_1T^c\rangle$ measured at the CG of the satellite transition instead of the central due to interference with the apical site. A dashed line indicates T_{scale}, and solid lines are taken directly from Fig. 2 with $\beta=2.4$ (sK%)$^{-1}$.

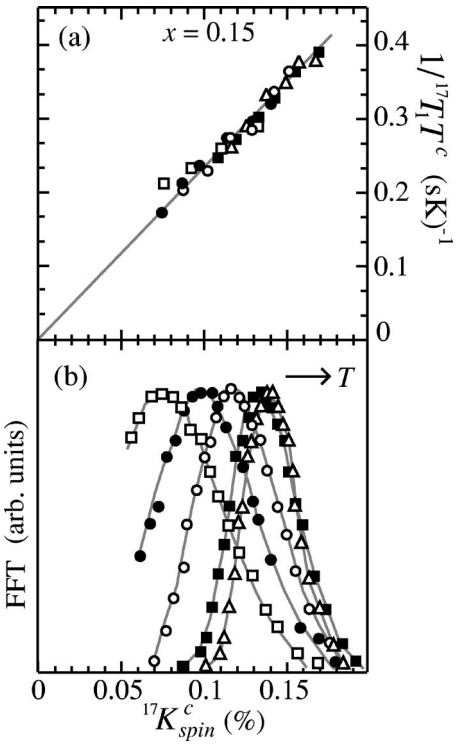


FIG. 5. (a) Frequency dependence of $1/^{17}T_1T^c$ (sK)$^{-1}$ for $x=0.15$ across ^{17}O NMR spectra $^{17}K^c_{spin}$ (%) presented in (b). A solid line in (a) is best fit to Eq. (1) with slope $\beta'=2.2(3)$ (sK%)$^{-1}$ (excluding 100 K data). (b) Planar ^{17}O FFT spectrum for $x=0.15$ at 366 K (△), 295 K (■), 230 K (○), 166 K (●), and 100 K (□). Missing data points in (b) indicate the region where an apical oxygen signal coincides. Solid lines in (b) are best fits using a patch model of the spatial variation $^{17}x_{local}$ with a patch radius $^{17}R_{patch}=2.0-4.4$ nm in order of increasing T, respectively.

findings for a c-axis aligned $x=0.15$ powders,[9] the temperature dependence of the linewidths show the same qualitative features.

In Fig. 4 we present the temperature dependence of $\langle 1/^{17}T_1T^c\rangle$ at the CG for all the samples. Equation (5) implies that $^{17}A(\mathbf{q})^{\alpha}\simeq(q_{\alpha}-\pi/a)^2$ in the vicinity of the in-plane antiferromagnetic wave vector $\mathbf{Q}_{AF}=(\pi/a,\pi/a)$. Therefore $1/^{17}T_1T^c$ is dominated by contributions away from the $\mathbf{Q}_{AF}$. From 475 K down to T_{scale}, where

$$T_{scale}=(300\text{ K},250\text{ K},175\text{ K},100\text{ K})\ (\pm 10\text{ K})$$

for

$$x=(0.035,0.05,0.115,0.15),$$

respectively, we find the same temperature dependence as $\langle ^{17}K^c_{spin}\rangle$ presented in Fig. 2. This is shown by the solid lines that are directly taken from Fig. 2, along with the $\beta=2.4(2)$ (sK%)$^{-1}$ value taken from Fig. 1.

Yet another way to illustrate the scaling relation in Eq. (1) is shown in Fig. 5, where we show the frequency dependence of $1/^{17}T_1T^c$ across the inhomogeneous broadening $^{17}K^c_{spin}$, however, instead of keeping T fixed and varying x, in Fig. 5 we keep $x=0.15$ fixed but vary T. The best fit linear extrapolation of the data points in Fig. 5 (excluding 100 K) results

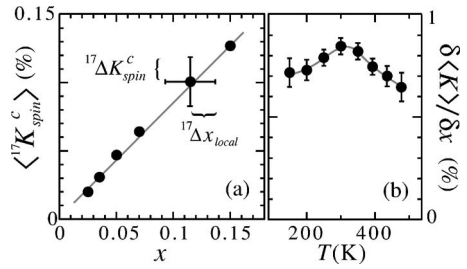


FIG. 6. (a) Spin shift at the CG $\langle ^{17}K^c_{spin}\rangle$ (●) against the mean local hole concentration, $\langle ^{17}x_{local}\rangle=x$, at 295 K. Also shown is the best fit first derivative (or slope) $\delta\langle K\rangle/\delta x$ (line), while the vertical error bars show $\pm^{17}\Delta K^c_{spin}$ with corresponding values of $\pm^{17}\Delta x^c_{local}$ (horizontal error bars) for $x=0.115$. (b) The temperature dependence of the slope $\delta\langle K\rangle/\delta x$ (●) in units of (%); the curve is a guide for the eye.

for the slope $\beta'=2.2(3)$ (sK%)$^{-1}$ and intercept $^{17}K'^c_{orb}=+0.010(8)$%, which are consistent with Fig. 1 to within experimental uncertainties.

B. Determination of $^{17}\Delta x_{local}$

As explained in the Introduction, the scaling relation of Eq. (1) across the spectrum of each crystal x (Fig. 1(a)) implies that the scaling relation is obeyed *locally*; therefore each value of the local shift $^{17}K^c_{spin}$ corresponds to a unique value of the local hole concentration $^{17}x_{local}$. In other words, there is a 1-to-1 correspondence between each local value of $^{17}K^c_{spin}$ and $^{17}x_{local}$. If, furthermore, a *linear* relation exists between the two, this implies a proportionality between the first moments $\langle ^{17}K^c_{spin}\rangle$ and $\langle ^{17}x_{local}\rangle=x$, which is given by

$$\langle ^{17}K^c_{spin}\rangle=\frac{\delta\langle K\rangle}{\delta x}x, \tag{6}$$

where we define $\delta\langle K\rangle/\delta x$ as the constant of proportionality. As presented in Fig. 6(a), we do indeed find a linear relation, which then implies that the *same* proportionality $\delta\langle K\rangle/\delta x$ exists between the second moments,

$$^{17}\Delta K^c_{spin}=\frac{\delta\langle K\rangle}{\delta x}{}^{17}\Delta x_{local}, \tag{7}$$

as shown by the error bars in Fig. 6(a).

Our discussion so far has been at a fixed temperature of 295 K, however, exactly the same procedure holds at different temperatures given that the scaling relation, Eq. (1), across each spectrum also holds at various temperatures (an example of this was shown in Fig. 5 for $x=0.15$). The only difference is that the proportionality constant $\delta\langle K\rangle/\delta x$ shows some temperature dependence, as presented in Fig. 6(b). Therefore, taking $\delta\langle K\rangle/\delta x$ from Fig. 6(b) together with $^{17}\Delta K^c_{spin}$ from Fig. 3, we determine $^{17}\Delta x_{local}$ using Eq. (7) at each temperature, and plot the results in Fig. 7. It is clear that $^{17}\Delta x_{local}$ increases with decreasing temperature below 475 K, which, according to our model in Sec. IV, is a consequence of a decrease in the characteristic length scale $^{17}R_{patch}$. Also, $^{17}\Delta x_{local}$ shows signs of saturation at high temperatures,

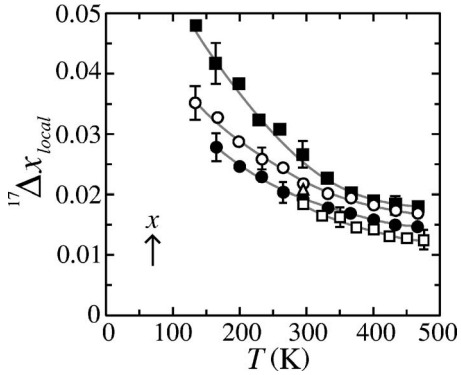


FIG. 7. Temperature dependence of the amplitude (or extent) of the spatial variation in local hole concentration, $^{17}\Delta x_{local}$, determined using the HWHM data in Fig. 3 above T_{scale}, and the conversion scheme illustrated in Fig. 6. Lines are a guide for the eye.

which for $x=0.15$ approaches its temperature independent EPMA value, $\Delta x_{epma} \simeq 0.012$. Δx_{epma} corresponds to the resolution limit for our technique, as illustrated by the shaded line shape in Fig. 1(b) in the case of $x=0.15$ (using the conversion in Eq. (7)).

We note that $^{63}\Delta x_{local}$ determined using the frequency dependence of $1/^{63}T_1$ across the ^{63}Cu NQR spectrum[12] shows semiquantitatively the same values as $^{17}\Delta x_{local}$, which adds weight to our whole analysis. However, the frequency dependence in $1/^{63}T_1$ does not necessarily imply a frequency dependence in $1/^{17}T_1$, and the fact that $^{17}\Delta x_{local} \simeq {}^{63}\Delta x_{local}$ has further consequences. $1/^{63}T_1$ probes the low frequency spin fluctuations near $\mathbf{q}=\mathbf{Q}_{AF}$, while $1/^{17}T_1$ probes them *away* from $\mathbf{Q}_{AF}$ at $\mathbf{q}\approx\mathbf{0}$. Similarly, $^{17}K_{spin}$ probes the static susceptibility *away* from $\mathbf{Q}_{AF}$, exactly at $\mathbf{q}=\mathbf{0}$. Furthermore, the line broadening of the ^{63}Cu NQR line over which $1/^{63}T_1$ varies is from the *charge* channel while the line broadening of the ^{17}O NMR line over which $1/^{17}T_1$ varies is from the *spin* channel; therefore our findings suggest a strong correlation between the spatial inhomogeneity in the charge and spin channels across most of the Brillouin zone.

IV. LENGTH SCALE ANALYSIS

We now turn our attention to a real-space static picture of the spatial variation $^{17}x_{local}$. Our *patch-by-patch* model is based on two simple ingredients: first, that the Sr^{2+} donor ions are randomly distributed in the lattice, and, second, that there exists a short characteristic length scale $^{17}R_{patch}$ (i.e., a patch radius) that describes the local electronic state in the random lattice. In our model, a patch represents an area in the CuO_2 plane through which there are a larger (smaller) number of itinerant holes passing through over time as a result of a larger (smaller) number of Sr^{2+} donor ions in the patch's immediate vicinity. This then results in $^{17}\Delta x_{local} > 0$ (<0) within that patch, respectively.

One can get insight into the physical significance of the patch model by taking a loose analogy with doped semiconductors, where the doped carriers are known to be influenced by the Coulomb potential of their donor ions. For low doping concentrations, this leads to an effective Bohr radius for the doped carriers around their donor ions. In our case, the higher concentration of *itinerant* doped holes are not perfectly screened from the nearby Coulomb potentials of the Sr^{2+} donor ions, which leads to a characteristic length scale of influence ($^{17}R_{patch}$) from the randomly positioned donor ions. In other words, if the itinerant holes were to perfectly screen each other from the donor ions, one would have $^{17}R_{patch}=\infty$ and hole doping would be homogeneous, however, as we have shown in Fig. 7, a substantial spatial variation in the local electronic state exists which can be naturally explained by a short $^{17}R_{patch}$.

We shall now describe the procedure for fitting the ^{17}O NMR spectra using the patch model in local electronic states, where the only free parameters in the fits are x and $^{17}R_{patch}$. We find that the optimized patch radius $^{17}R_{patch}=2-5$ nm is consistent with previous estimates of $^{63}R_{patch}$[12] according to a similar analysis for ^{63}Cu NQR spectra. We note that the appropriate "static" time scale of the NMR spectrum, and therefore of the real-space model, is given by $2\tau \sim 20$ μs, where τ is the experimentally chosen excitation-pulse separation time.[33] $1/2\tau \sim 5\times 10^4$ Hz is the effective shutter speed of the NMR picture, and therefore represents a static picture of the itinerant holes.

A. Procedure

We first assume that the Sr^{2+} donor ions are randomly distributed within the La^{3+} sites; therefore the probability of a Sr^{2+} ion at a particular La^{3+} site is given by $x/2$. Given that there are two La^{3+} sites per CuO_2 (one above and one below the plane), there is then a probability of x for one hole donation per CuO_2. For simplicity, we neglect the small probability ($x^2/2$) of double hole donation per CuO_2 since we are in the range $x \leqslant 0.15$. Once this is carried out for each CuO_2 site, we have achieved the particular random configuration κ, as illustrated in Fig. 8. We exclude the possibility of any Sr^{2+}—Sr^{2+} clustering beyond probability theory, which is supported by the absence of any significant diffuse scattering in neutron diffraction experiments.[45]

Taking the origin at a Cu site for convenience, we then ascribe a patch of radius R_{patch} within the CuO_2 plane that covers a certain number N^R of Cu sites, or $2N^R$ oxygen sites equivalently, where $N^R=\pi R_{patch}^2/a^2$, an example of which is shown in Fig. 8 for $x=0.115$ and $R_{patch}/a=8$. We then count the random number of donated holes N_{Sr}^{κ} that are uniformly shared among the N^R CuO_2 sites within the radius R_{patch}. Finally, we determine the patch's hole concentration x_{patch}^{κ} as

$$x_{patch}^{\kappa} = N_{\mathrm{Sr}}^{\kappa}/N^R. \tag{8}$$

We then randomly recreate the configuration $\kappa \sim 10^4$ times, and count x_{patch}^{κ} for each configuration. The histogram of x_{patch}^{κ} is then converted to a histogram in $^{17}K_{spin}^{c}$ by multiplying by $\delta\langle K\rangle/\delta x$ from Fig. 6(b).

The resulting best fits to the ^{17}O NMR line shapes are shown in Fig. 1(b) for 295 K with the corresponding optimized values of R_{patch}, defined as $^{17}R_{patch}$, presented in Fig. 9. We find that $^{17}R_{patch}$ shows a mild $\pm 10\%$ x dependence at 295 K, and the drop in $^{17}R_{patch}$ for $x=0.035$ is due to the fact that this sample is already in the temperature region where

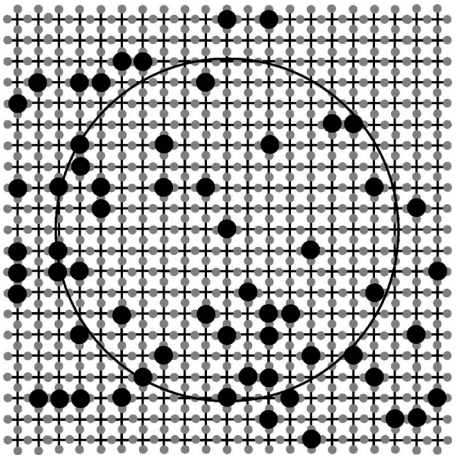


FIG. 8. An illustration of a CuO_2 plane where the Sr^{2+} ions (●) are randomly positioned in the lattice with $x=0.115$. The corners of each square represent a Cu site, while the planar oxygen sites are shown as (grey ●). Also shown is a typical patch (circle) with radius $R_{patch}=3.0$ nm ($=8a$) for which this particular random configuration κ yields $x^{\kappa}_{patch}=0.139$.

the scaling relation breaks down, and the validity of our model is questionable in this region. We note that our simulation shows that for a given x and a given temperature, a *single value* of $^{17}R_{patch}$ best fits the line shape. In other words, adding any distribution to $^{17}R_{patch}$ across a given CuO_2 plane only adds weight to the tail of the spectrum, which is inconsistent with experiment.

In Fig. 9 we also plot the *lower bound* $^{63}R_{patch}$ (i.e., the optimized value of R_{patch} according to ^{63}Cu NQR data[12]) for which a consistent value to the length scale is found, which adds weight to our model. We also show the average nearest neighbor Sr^{2+} distance l_{Sr} according to numerical simulation, which fit well to the form $l_{Sr}\simeq 0.56a/\sqrt{x}$ in the range of x

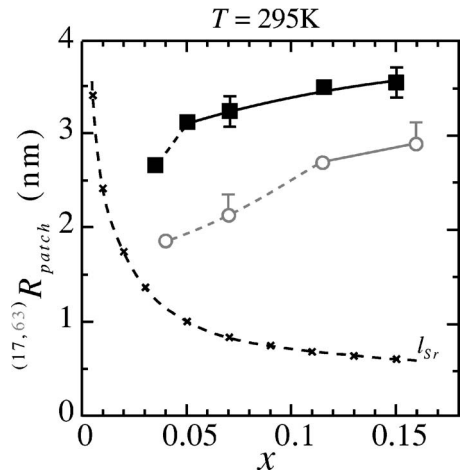


FIG. 9. Best fit values of $^{17}R_{patch}$ (nm) (■) used to generate line shapes in Fig. 1(b) at 295 K, where the solid (dashed) line indicates the region above (below) T_{scale}, respectively. For comparison, the *lower bound* $^{63}R_{patch}$ (nm) (grey ○) according to ^{63}Cu NQR[12] at 295 K is shown, where the grey solid (dashed) line shows the tetragonal (orthorhombic) structural phase (Ref. 46), respectively. Also shown is the average nearest neighbor Sr^{2+} distance (×).

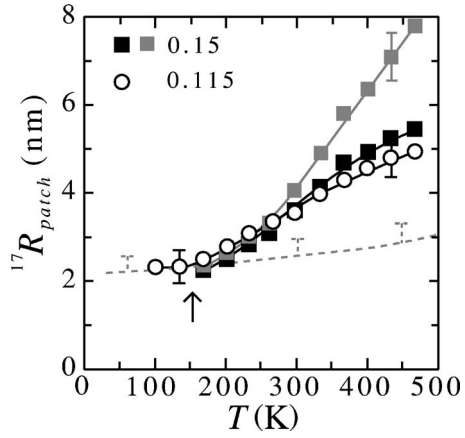


FIG. 10. Temperature dependence of best fit values $^{17}R_{patch}$ (nm) for $x=0.15$ (■) used to generate line shapes in Fig. 5(b), together with $^{17}R_{patch}$ for $x=0.115$ (○), and arrow indicates T_{scale}. Also shown is $^{17}R'_{patch}$ for $x=0.15$ (grey ■) according to Eq. (11). Solid lines are a guide for the eye, while a dashed line indicates the lower bound $^{63}R_{patch}$ for $x=0.115$ taken from Ref. 12.

shown. The fact that l_{Sr} ($\ll{}^{(17,63)}R_{patch}$) implies that each patch encompasses many Sr^{2+} sites.

Finally, we repeat the whole fitting procedure at different temperatures, and the resulting best fits to the ^{17}O NMR line shapes are shown in Fig. 5(b) for $x=0.15$ with the corresponding optimized values $^{17}R_{patch}$ presented in Fig. 10. We find a clear decrease in $^{17}R_{patch}$ with decreasing temperature for all x.

B. Discussion

Our simulation shows that a simple relation exists between the $^{17}\Delta x_{local}$ data in Fig. 7 and the optimized values of $^{17}R_{patch}$ we have deduced to fit the line shapes. According to our numerical simulation, the calculated second moment of the distribution $(\Delta x)^2$ is related to the free parameters x and N^R as such,

$$(\Delta x)^2 = \frac{x(1-x)}{N^R}. \tag{9}$$

Equation (9) resembles the second moment of a binomial distribution, which is not surprising since our simulation is nothing more than a example of a binomial distribution with probability x over sampling number N^R.[47] Substituting $N^R = \pi R^2_{patch}/a^2$ and using the current limit $x/2 \ll 1$, Eq. (9) reduces to the HWHM,[42]

$$\Delta x \simeq \frac{1.18a}{\sqrt{\pi}}\frac{\sqrt{x}}{R_{patch}}, \tag{10}$$

where all of the temperature dependence of Δx originates from R_{patch}. The relation in Eq. (10) can be verified by comparing the $^{17}\Delta x_{local}$ data in Fig. 7 to the optimized values of $^{17}R_{patch}$ in Figs. 9 and 10.

A refinement to our calculation of $^{17}R_{patch}$ can be made for $x=0.15$ sample if we take into account the temperature-independent EPMA value of Δx_{epma} as an additional source

of line broadening. Given that Δx_{epma} and Δx from Eq. (10) are independent broadening mechanisms, we can approximate the total observed width $\Delta x'$ as

$$\Delta x' \simeq \sqrt{(\Delta x_{epma})^2 + (\Delta x)^2}, \tag{11}$$

where $\Delta x_{epma} \simeq 0.012$, and Δx is given in Eq. (10). Equating $\Delta x'$ in Eq. (11) to the experimentally observed width $^{17}\Delta x_{local}$ results in a different deduced value $^{17}R'_{patch}$, shown in Fig. 10 as the grey squares. At low temperatures $^{17}R'_{patch} \simeq {}^{17}R_{patch}$, but at high temperatures $^{17}R'_{patch} > {}^{17}R_{patch}$, where the observed line broadening becomes dominated by Δx_{epma}. Regardless of whether one includes the EPMA result in the model, $^{17}R_{patch}$ is clearly found to increase above T_{scale}, which within our model implies that hole doping becomes more homogeneous with increasing temperature.

We note that similar patch-by-patch models of disordered doping in the CuO_2 plane have been used by Martin and Balatsky[48] to account for the smeared density of states found in photoemission peaks as a result of a distribution in the superconducting gap. The free parameter in their model is the length scale that they attribute to the in-plane superconducting coherence length ξ_{ab}. It is then interesting to note that our electronic length scale $2R_{patch} \sim 4$ nm in the low-temperature region (Fig. 10) is comparable to $\xi_{ab} \sim 3.3$ nm observed in thin films.[49]

V. CONCLUSIONS

We have presented a systematic study of the spatial variation in the electronic states using ^{17}O NMR in high-quality single crystals of $La_{2-x}Sr_xCuO_4$ for $0.035 \leqslant x \leqslant 0.15$ from three different sources. We found that the data can be consistently accounted for by alloying effects over a short characteristic length scale $^{17}R_{patch} = 2-5$ nm. We also show consistency between the ^{17}O NMR results and the ^{63}Cu NQR[12,25] results, indicating that the spatial variation of the spin and charge degrees of freedom are correlated through the local hole concentration. Our findings have clear implications on the general, but now incorrect view, that hole doping is homogeneous in the CuO_2 planes of $La_{2-x}Sr_xCuO_4$. We note that very recent *ab initio* calculations in $HgBa_2CuO_{4+\delta}$ have revealed pronounced intrinsic inhomogeneity effects on hole doping, electric field gradients, and density of states,[50] which are entirely consistent with our ^{63}Cu NQR[12,25] and ^{17}O NMR (present work) experimental findings in $La_{2-x}Sr_xCuO_4$.

We also note that the presence of the short length scale also implies that any bulk probe that averages over length scales larger than $2R_{patch} \sim 4-10$ nm results in an *underestimate* for the extent Δx_{local} of the spatial variation. NMR and NQR, however, are local probes making them ideal for detecting ~nm spatial variations in the electronic state over the bulk of the material.

The scaling between the spin dynamics and the spin shift is eventually broken below the temperature region T_{scale}. Below T_{scale}, we find that $1/{}^{17}T_1T^c$ saturates (Fig. 4) while $^{17}K^c_{spin}$ continues to decrease (Fig. 2) with decreasing temperature. We also find that the temperature dependence of the linewidth (Fig. 3) saturates below T_{scale}. One possible reason for the departure from scaling is that $1/{}^{17}T_1T^c$ picks up contributions from growing *incommensurate* spin fluctuations[51] at the NMR frequency near $\mathbf{q} \approx \mathbf{Q}_{AF}$.[36] The presence of finite incommensurability implies that the hyperfine form factor in Eq. (5) does not entirely filter out the AF fluctuations, thereby causing an enhancement in $1/{}^{17}T_1T^c$ over the uniform susceptibility $^{17}K^c_{spin}$, thus a departure from scaling. Another possible reason for the departure from scaling is that for $T \leqslant T_{scale}$ there are a growing number of regions where $^{17}x_{local}$ is very small. We see this in Fig. 7, where at T_{scale}, $^{17}x_{local} = x - {}^{17}\Delta x_{local}$ for the insulating regions get closer and closer to zero, implying that certain regions in the CuO_2 plane are close to being undoped. Such regions may behave as free-spin-like entities that enhance $1/{}^{17}T_1T^c$ for neighboring sites, as well as cause the wipeout of the ^{63}Cu NQR intensity for $x < 0.115$.[52]

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